

Class-conditional sourcing

two classes, two collection routes

counterfeit = screenshot
authentic = photo

Two acquisition pipelines

format, size and lighting differ by class

PNG vs JPEG
405 vs 223 px
339 vs 6 kB

Acquisition predicts the label

$I(Y; A) > 0$; complete when $H(Y | A) = 0$

format alone: 510/510
header model: 1.000

Every partition inherits it

stratified and grouped alike

9/480 groups leak
0.3 pt varied alone

detectable here,
before any model is trained

High in-distribution accuracy

cannot separate packaging from provenance

0.919 on the leakage-free test set

A new acquisition pipeline

images the authors did not collect

150 external authentic photographs (condition C)

Failure

the shortcut is absent

9/150 correct (6.0%)